Towards a demographic analysis of kinship bereavement

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Contents

Abstract

The death of a family member can have significant implications for an individual's well-being. Different studies have explored and quantified the prevalence of specific types of bereavement or focused on the long-term impact from a life-course perspective. In societies that have undergone demographic transition, kin death has become less common due to smaller kinship networks

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and lower risks of dying. In this context, it is necessary to evaluate and compare distinct bereavement process from a formal demographic perspective. We propose a series of measures to assess the extent and timing of bereavement within a kinship network, as experienced by an average individual over age in a theoretical population, in the spirit of period measures like life expectancy. These indicators are focused on duration (Expected loss of shared kin-time, Mean time since experienced loss) and prevalence of specific situations associated with severe grief (Non-expected proportion of kin dead, Overlapping generational sequence). From a general perspective, this reflects the implicit bereavement conditions stemming not from a specific shock or change, but from the overall levels and patterns of mortality and fertility across different age groups. As an example of some results, an Argentinian woman with 60 years old, living under 2015 year conditions, would expect to live 16.5 years without her mother (spending the last quarter of her life in bereavement stage). This is extended to various ages and types of kin, considering also the rest of dimensions. We present findings from Argentina and Guatemala, contrasting the population's bereavement experience during various historical demographic periods. Kinship and bereavement are universal, so demographic analysis can model an average synthetic behaviour and get the implicit patterns. The next question is which ones?

1 Background

The loss of kin is a universally experienced life event (Shear, 2012), which comes from the fact that kinship is universal (Caswell, 2019), and mortality is the only non-avoidable demographic component. More than 60 million annual deaths worldwide (United Nations, Department of Economic and Social Affairs, Population Division, 2022) leaves behind a kin network, recently measured at 9 in average for United States of America (Verdery et al., 2020). Each of those relatives transits to the state of bereavement at a certain age, given a certain kin type of the deceased, within a kinship structure that evolved along his/her existence and is expected to continue changing during their lifespan. From a population-level perspective, given a fixed regime of fertility and mortality, this creates a specific life-course average experience of bereavement. From a health-policymaker perspective, knowing these characteristics is important for planning interventions in a life-course event that can be associated with increasing short-term and long-term mortality (Prior et al., 2018; Colin Murray and Holly G., 2009). Recent public-health protocols has been developed to guide bereavement risk assessment, identifying groups with highest risk of mental illness and having a medical support (Aoun et al., 2015). Grief stage (the first stage in bereavement process) is related with a bad self-reported health, mental health issues and a more intensive use of health services (Thimm et al., 2020). Also how close in terms of kin, to the deceased is, means a significant factor to predict grief severity (Lobb et al., 2010).

For the purpose of clarifying key terms along this work, we refer to bereavement as the "experience of having lost someone close", which turns an individual to bereaved, staying in that state over the lifespan. To grief as the "psycho-biological response to bereavement", while mourning to the "array of psychological processes that are set in motion by bereavement in order to moderate and integrate grief" (Shear, 2012). With kin we refer strictly to a co-sanguinely definition. Based on recent methodological advances in kinship network estimations (Caswell, 2019), we can explore specific dimensions of the bereavement process which can help to compare populations in time and space, studying how demographic changes over time transforms the average bereavement experience by age.

2 Methodology

2.1 Kinship Demography

Our results have as its main input the expected distribution of living and death relatives by age of Focal (or Ego), type, age and sex of the kin. All results that comes from the kinship estimation methods in a female-dominant, 2-sex and time-invariant framework, implied by a set of rates of fertility and survival probabilities in a theoretical population (Caswell and Song, 2021). This has the same nature of period-type summary indicators as life expectancy at different ages from life tables. All calculations assume independence between the survival of Focal and her relatives, and also between relatives.

We will work with two main functions. $k(x)$ is the number of living kin of type k when Focal is x years old. For convenience we will generalize the GKP formulas (Goodman, 1974), splitting $k(x)$ in two parts (eq. 1). A first term of living kin that preexists Focal, with an initial distribution $\alpha(z)$ by age z when Focal born. For example the parents starts being 2 (if maternal and paternal mortality during pregnant is not taken), so $\int_0^\omega \alpha(z)\,dz = 2$. Those needs to survive to x for being alive at Focal´s age x . The second part of $k(x)$ is given by $\beta(x)$, which represents the subsidy or recruitment term: newborns that comes from Focal's reproduction or from other type of reproduction (for example, grandchildren coming from children), which depending on where those are born in Focal's lifeline, also needs to survive until Focal is aged x to be counted in $k(x)$. An assumption is implicit that new kin given by β do not consider, for example, adoptions where the age of entry to Focal's network can be greater than 0 (co-sanguinely definition).

$$k(x) = \int_0^\omega \alpha(z)\frac{l(z+x)}{l(z)}dz + \int_0^x \beta(y)\frac{l(x-y)}{l(0)}dy \tag{1}$$

Here $l(x)$ is the survival function from the life table. As a reference for each part consider: for mothers $\beta(y) = 0$ (no new mothers after Focal born), for daughters $\alpha(z) = 0$ (there are no children when Focal born). This means that, depending of younger or older cohorts between Focal and her relative, living kin of age z when Focal is x is $k(x,z) = \alpha(z-x)\frac{l(z)}{l(z-x)}$ if $z > x$ or $\beta(y)\frac{l(x-y)}{l(0)}$ when $z < x$. Total living kin at certain Focal age comes from summing over kin age distribution: $k(x) = \int_0^\omega k(x,z)\,dz$.

The other important measure is $d(x)$, which is the number of deaths of k type when Focal is x yo, which considering kin age is $d(x,z) = \int_0^\omega k(x,z)\,\mu(z)dz$, being $\mu(z)$ the mortality rate at age z. Measures based in he effect of a shock in the kinship tree deals with this last expression, evaluating elasticity on $\mu(z)$. Previous expressions are enough for building the core content of this work.

In this context, heterogeneity can be studied from an individual perspective (by chance) or comparing between groups. We compare Argentina and Guatemala along the work, two countries that had very different demographic transition pace along last 70 years within the Latin American context (8): a life expectancy at birth from 60 to 76 in Argentina, and from 40 to 73 in Guatemala, a TFR from 3.1 to below 2 in Argentina, and from 7 to 2.5 in Guatemala. Mean age at death (ages greater than 10) presents a constant absolute difference of 15 years (main convergence in level comes from reducing infant mortality in Guatemala), and mean age at childbearing shows a youngest age for Argentina during almost all the period. Expected kin networks depends fully of level and patterns by age and sex of mortality and fertility. Distribution of parents, grandparents and children in 1950 and 2015 shows the expected death experience from a period perspective for a new born given the demographic natural regime at that year (see 1): in 1950 Guatemalan case has more dispersion by age for all types, with a first mode by age at death for children and siblings, influenced by high infant mortality. When accumulating deaths by age of focal, at 30 yo an Argentinian in 1950 would lost have .6 parents and a Guatemalan 1 (with more dispersion by age), would lost .24 children (from 2.6 already had, ~10%) compared to 1.5 (from 5.2, ~30%). At 2015 there is a very similar pattern and level by age for all types considered, reflecting transition in 8.

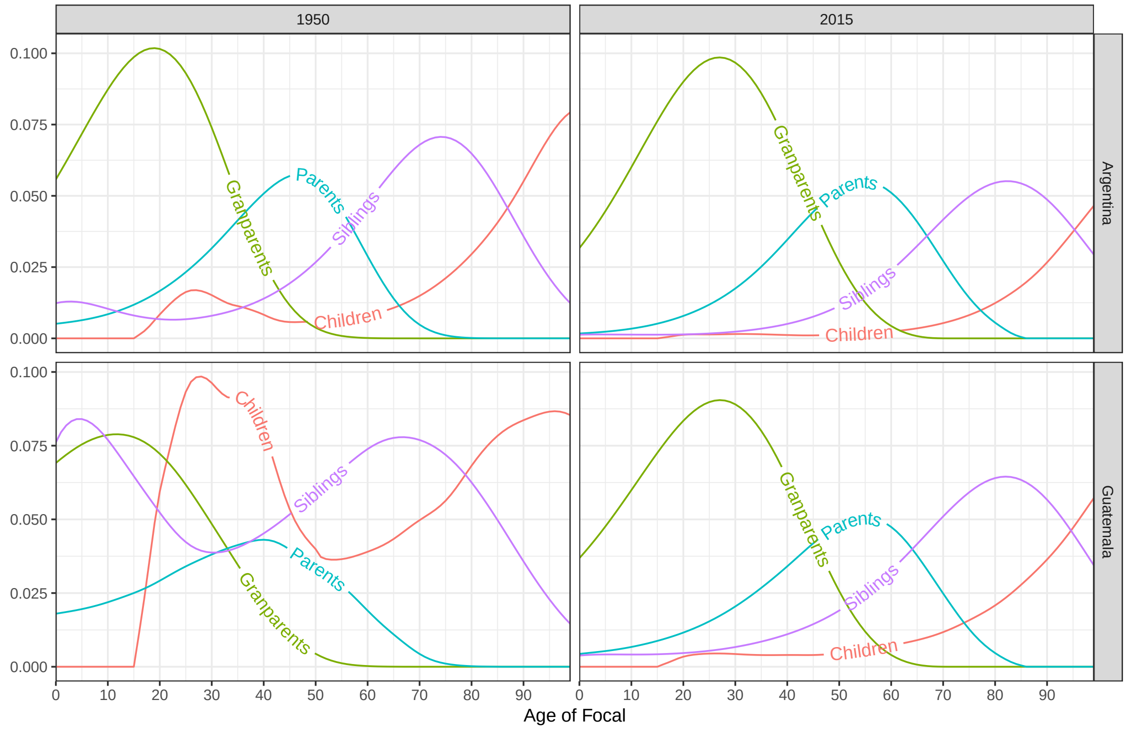

Figure 1: Death experience over age for a newborn from Argentina and Guatemala in 1950 and 2015, for selected kin types, in a time-invariant scenario.

Now is convenient to explore how demographic analysis can characterize this process beyond expected count distribution.

2.2 How to measure demographic bereavement?

How can the bereavement process be measured from an analytical demographic perspective? This question does not have a straightforward answer, primarily due to the multifaceted dimensions involved. These dimensions include the ages of both the focal individual and the relative, their respective genders, the type of kin relationship, whether the perspective is prospective or retrospective (considering the burden of past experiences or anticipated loss), as well as cultural and socioeconomic factors, among others. Here we focus in a set of indicators that covers some aspects where there is a proven relationship between bereavement and potential grief factors: Expected loss of shared kin-time (S), Unexpected proportion of kin dead (U) and Overlapping generational sequence (O). Each measure is defined for any type k (a single kin type or a group), being more relevant for some types than others in each case: for a woman aged 60 years old, it seems more interesting to know the expected time in bereavement because of a child loss than for a parent loss.

The goal is to give a broader picture of the loss experience in a kinship network for an averaged person at each age, given a set of survival and fertility risk pattern by age. Assume Focal is 60 years old in Argentina 2015; we can say about her daughters: Focal had 2.3 kin ever met (no more will be born), from which 2% died, being in bereavement the last third of her life, being the 84% from an unexpected death. For the rest of her life ($e(60) = 24$), she will experience 0.13 deaths, expects to be 6.5 shareable years in lost.

In the next sub-sections, details about meaningful and construction of the indicators are given. Others are detailed in the section 5.3, some being used along the work. In naming with letters the types of kin, we will refer to the female dominant version, although we are treating both sex: m (mother) for parents, d (daughter) for children, etc.

2.3 Expected loss of shared years

What is the expected time of child loss when Focal is aged 30? This will depend on the fertility and mortality level and pattern by age: more offspring at younger ages, with children dying fast, means more expected kin-years of loss (in analogy with $e^{\dagger}$ from Vaupel et al. (1979)), but this time for a population: kin are populations, so a multi-cohort approach is needed. Let's see a simple intuitive example.

In figure 2 Focal is 30 years old. If we know that her mother, aged 50, will die at age 80 (later in this work this will be an entire age distribution), then the expected time in bereavement will be $\frac{l(60)}{l(30)}\, e(60)$. Her grandmother already died when Focal was 10 years old, so the expected time is $e(30)$. For daughters, we have three different situations: one offspring died five years before, the other is living (and will share all Focal's life), and two else will be born in 5 and 10 years, the first dying 10 years after, and the second will survive Focal's death. For the first death, time expected is $e(30)$, as in the grandmother's case; this is a past certain death. For the second, it will be $\frac{l(45)}{l(30)}\, e(45)$. What is the total expected time in bereavement for daughters when Focal is aged 30? For this simple case, it would be $e(30) + \frac{l(45)}{l(30)}\, e(45)$, which is greater than $e(30)$ in any human society. So we need an extra step to find a final interpretation of this measure. Here we introduce the idea of kin-years: amount of expected time in bereavement state from a certain age. In the next paragraphs we will treat expected values, with a probability distribution for fertility and mortality, we won't treat individual lines like in this example.

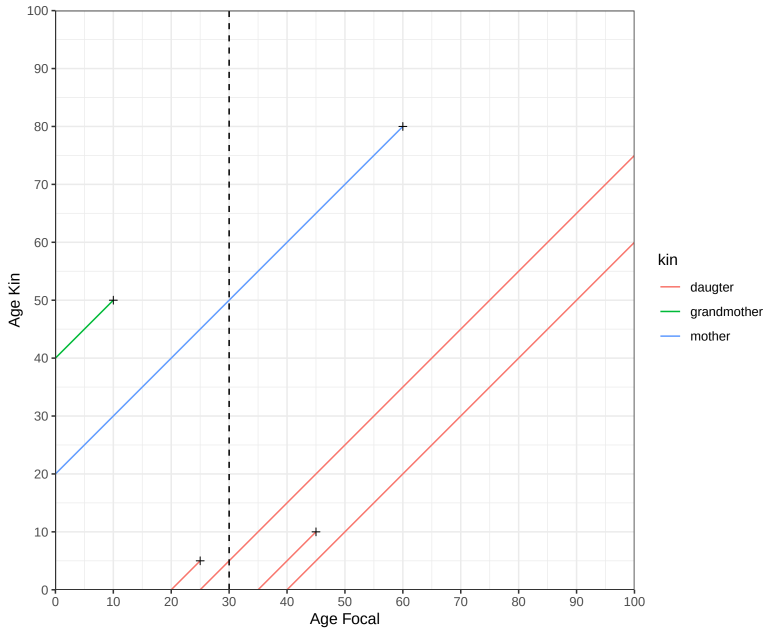

Figure 2: Unique and multiple types of kin. The expected time in bereavement needs a population-cohort approach, counting kin-years.

Let's generalize the case for any age of Focal x, any type of kin k. The expected time in bereavement at age x can be decomposed into past experience (accumulated in younger ages), future experience from the living stock, and future experience from new kin, that will be born (recruited or subsidized in general) in the future ("future" means Focal growing). For the first part, Focal will transit all her expected future years in that state:

$$\mathsf{T1}(x) = D(x)\, e(x) \tag{2}$$

Where $D(x) = \int_0^x d(y)\, dy$ is the cumulative deaths of kin k for a Focal aged x. That is the case of the grandmother in figure 2.

For the living kin at age x the distribution by age of death for each kin cohort, times the life expectancy of Focal at that age, is the total expected time. Here, $\frac{l(x+t)}{l(x)} e(x+t)$ is a deferred life expectancy (the same concept introduced in figure 2 for the mother), of Focal surviving to the exact time her kin dies. But also Focal will experiment deaths from kin generations born during her life, at Focal's age y ($y > x$), that will die with z ($z < x$) years old. Identifying this is relevant since age of deceased is inversely correlated with grief severity (Aoun et al., 2015), so this is why we introduced equation 1. This two groups (those alive when Focal born and those not yet) will contribute to the expected kin-years $\mathsf{T2}$:

$$
\begin{aligned}
\mathsf{T2}(x) &= \int_x^\infty \frac{l(a)}{l(x)}\, e(a)\, d(a)\, da \\
&= \int_x^\infty \frac{l(a)}{l(x)}\, e(a)\, [\int_0^\infty d(a,z)\, dz]\, da \\
&= \int_x^\infty \frac{l(a)}{l(x)}\, e(a)\, [\int_0^{a-x} d(a,z)\, dz + \int_{a-x}^\infty d(a,z)\, dz]\, da
\end{aligned} \tag{3}
$$

The amount of expected time in years, $\mathsf{T}(x)$, is the sum of *kin-years* in bereavement because of past, present, and future kin deaths. In a general expression:

$$
\mathsf{T}(x) = \mathsf{T1}(x) + \mathsf{T2}(x) \tag{4}
$$

But what is the mean time *shared* that Focal would lose? With sharing, we mean that Focal and her relative needs to be alive at the same time. In actuarial literature, joint survivorship theory treats the survival of two individuals as two separate random variables: if $T(x)$ and $T(y)$ are time to death for two persons aged x and y, then $T(y,x) = \min[T(x), T(y)]$ is the time distribution where the first death occurs, and $p(y,x)$ is the joint-survival distribution associated. Assuming independence between risks, risk rate is the sum of both and we have that $p(y \cap x) = p(x)p(y)$, with the expected time where both are alive as $e(x \cap y)$ (Bowers et al., 1997; POLLARD, 1991). The theory of copulas tries to relax this assumption modelling the joint distribution based on marginal distributions and correlation by age. Family correlation on survival requires observed data, which means a set of registered ages at death for relatives, father and son for example, which can be found for example in genealogical databases (Cabrignac et al., 2020).

In analogy with $\mathsf{T1}(x)$, given a dead relative, the expected shared time is zero ($\mathsf{S1}(x) = 0$), the relative is already dead. The computation for $\mathsf{S}(x)$ is the same as $\mathsf{T2}(x)$ but replacing Focal's life expectancy with joint-life expectancy, now considering kin's death age too: $e(x \cap z)$. The joint life expectancy only needs Focal to survive, because the expected occurrence of kin death assumes the survivorship until that age. We can separate the expected kin-years from present and "future" generations too (second line):

$$
\begin{aligned}
S(x) &= \int_x^\infty \frac{l(a)}{l(x)} \int_0^\infty d(a,z)\, e(a \cap z)\, dz\, da \\
&= \int_x^\infty \frac{l(a)}{l(x)}\, [\int_0^a d(a,z)\, e(a \cap z)\, dz\, + \int_a^\infty d(a,z)\, e(a \cap z)\, dz] da
\end{aligned} \tag{5}
$$

As $\mathsf{T}(x)$, this is a prospective indicator, measured in kin-years of shared-time, by definition $\mathsf{S}(x) < \mathsf{T}(x)$. This measures *kin-years* (an amount of *time*), not life length, and is possible to be greater than $e(x)$. Compared with the synthetic cohort approach used in life table methods, the exposures can increase in number with age. Many possibilities arise when interpreting this value in terms of $e(x)$, the mean life length of Focal. A possible interpretation is how many lives can expect to spend in bereavement because of kin type k, $\frac{\mathsf{S}(x)}{e(x)}$. Another possible way to interpret this is in terms of Focal's lifespan, which cannot be greater than $e(x)$, so $\min[e(x), \mathsf{S}(x)]$. A third one is the mean time for each ever-known kin, $\frac{\mathsf{S}(x)}{L(x)}$, which refers to how many years of bereavement have per each relative she has, and is more neutral in terms of fertility level. We will continue this exploration in the application section, but certainly this depends of the type of question to answer.

2.4 Unexpected dead

What is an *unexpected* death and why does it matter? The definition is not unique. From an epidemiological perspective, an unexpected death of some relative can be defined by its cause, specific at an epidemiological transition stage. External causes like suicide carry a specific grief process with high psychological impact Pitman et al. (2016). Non-life limiting illness and violent deaths are a relevant factor for higher risk of Prolonged Grief Disorder (Aoun et al., 2015; Lobb et al., 2010). A matrix implementation of kin loss distribution by cause is proposed in (Caswell et al., 2023). From a life table approach is possible to define a arbitrary threshold age, or an age related to the mature component of total death distribution by age Mazzuco et al. (2021). Youth (those aged ¡18 years) parental death has been associated with negative health outcomes Schlüter et al. (2024). Because time availability of cause decomposition, for the sake of illustration, in the application section this will be the age where dies less than 25% of the synthetic cohort in the period life table.

The calculation is simple: define $D^u(x)$ as the cumulative deaths of kin type k until Focal age x, that are classified as unexpected and get the proportion over total accumulated deaths:

$$\mathsf{U}(x) = D^u(x)/D(x) \tag{6}$$

2.5 Overlapping experience

Bereavement kin experience is oriented vertically by degree: Focal expects her grandmother to die before her mother, her mother to die before her children and siblings, her cousins before her aunts, and so on. A disrupted "natural" order is an important factor for traumatic grief process, like the child loss case Alburez-Gutierrez et al. (2021). Given a mortality function by age, we can think in the probability that parents dies before the grandparents, $P(T_{\text{gd}} > T_{\text{d}})$, with two random variables T_{p} and T_{gp}, ages at death. But the complexity arises when we have a *population* in each kin type, which can grow with Focal´s age, and there are infinite possibilities of measures (dies all, the first, etc). Another factor of complexity is that we have an expected age distribution at each time for each kin type. To keep it simple and valid for any possible pair of types, we can define an indicator that takes the share of total deaths where both types occurs, overlapping the expected frequency. Even when Focal expects she dies before her children, at older ages an unnatural death would be of their grandchildren rather than children. When a grandchild dies, grandparents experience a dual sorrow: cry for their lost grandchild, and struggle with wanting to help their adult children (Youngblut and Brooten, 2018).

This is an indicator that relies on more than one type. If j is the type which is expected to die before k, then the portion of bereavement experienced that is *overlapped* until age x is:

$$\mathsf{O}(x) = \frac{\int_0^x \min\{d_k(x), d_j(x)\}\, dx}{\int_0^x [d_k(x) + d_j(x)]\, dx} \tag{7}$$

Consider the case of the expected overlapping experience with grandparents related to parents in 3. When grey area is bigger, hierarchical order is less common. The maximum is always in the age where the same amount of deaths are produced between related kin types. The chances of overlapping experiences decrease with lower fertility and survival dispersion (lower $e^\dagger$ in the last case).

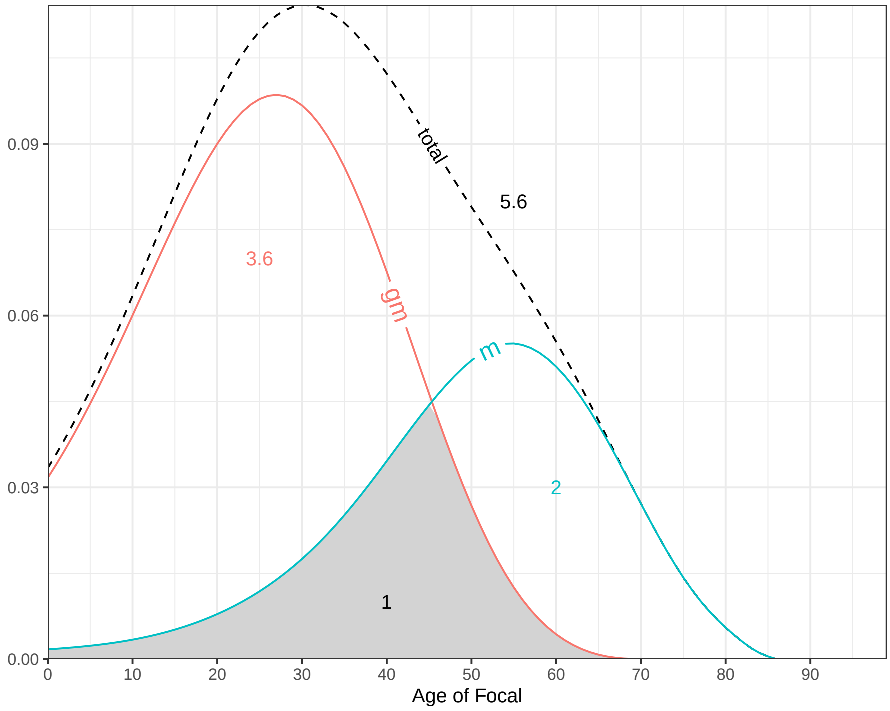

Figure 3: Parents and grandparents death counts along Focal´s life. Grey area gives the mother death experience that overlaps with grandmother experience. Indicator $O(x)$ refers to the proportion of cumulative grey area over total, until age x. In this example, when Focal is aged 90, $O(90)$=1/5.6=.18. A 18% of her death experience on parents and grandparents was overlapped.

The criteria about how to bi-relate types is arbitrary, but for a general rule schema see 4, which seems a reasonable guide, and will be used in the section 3.

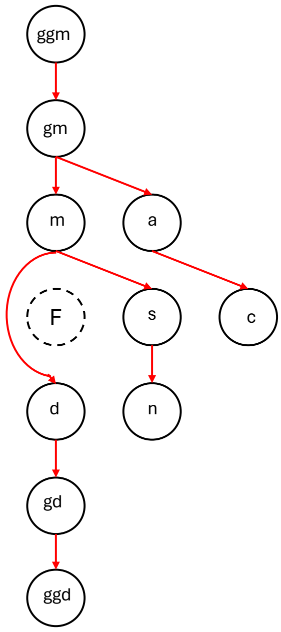

Figure 4: Theoretical schema of "non disrupted" order of death experience by kin type. Grand-granddaughters dying first than great-granddaughters, daughters before granddaughters, and so on.

3 Bereavement patterns over time

3.1 Argentina in 2015

An Argentinian woman living under 2015 conditions would experience 27 deaths throughout her full life (assuming she lives to age 100), a third of them before age 60 and half by age 72. The frequency of kin deaths by age for main types is shown in figure 1, in top-right panel: deaths are higher for gm until age 45, for parents until age 65 and then for siblings. The modal age for losing a grandparent was between 25 and 30, for a parent was around 55, and between 80 and 85 for siblings.

Let´s take a look to bereavement pattern over age for a woman. A brief picture of results are in table 1. When Focal born (e_0=80.6) she has 2 living parents, and expects to spend 20.8 kin-years ($S(0)$, see fig 5) of shared time without parents (which means that if mortality by sex is the same then she would spend 10.4 because each one, mother and father). When she is 30 yo, she expects to have passed a third of her life if there was a dead parent ($M(x)$), 3% of which were unexpected assuming the age of 30 as threshold, and 8% of her losses of parents and grandparents were overlapped. In later life, when she is 60 yo, the overlapping experience would increase to 18%, given the increase in death parents in last 30 years of life. For children, Focal at age 60 would expect to have 2.2 living kin, with 3% already dead, expecting to pass 2.3 kin-years of shared life (compared to 24.4 of expected life, 9.4%), with 85% of unexpected deaths. Same analysis can be done with other types: when Focal born she expects to live 34.6 kin-years without sharing time with grandparents, and a 13% of her siblings deaths as unexpected at age of 60.

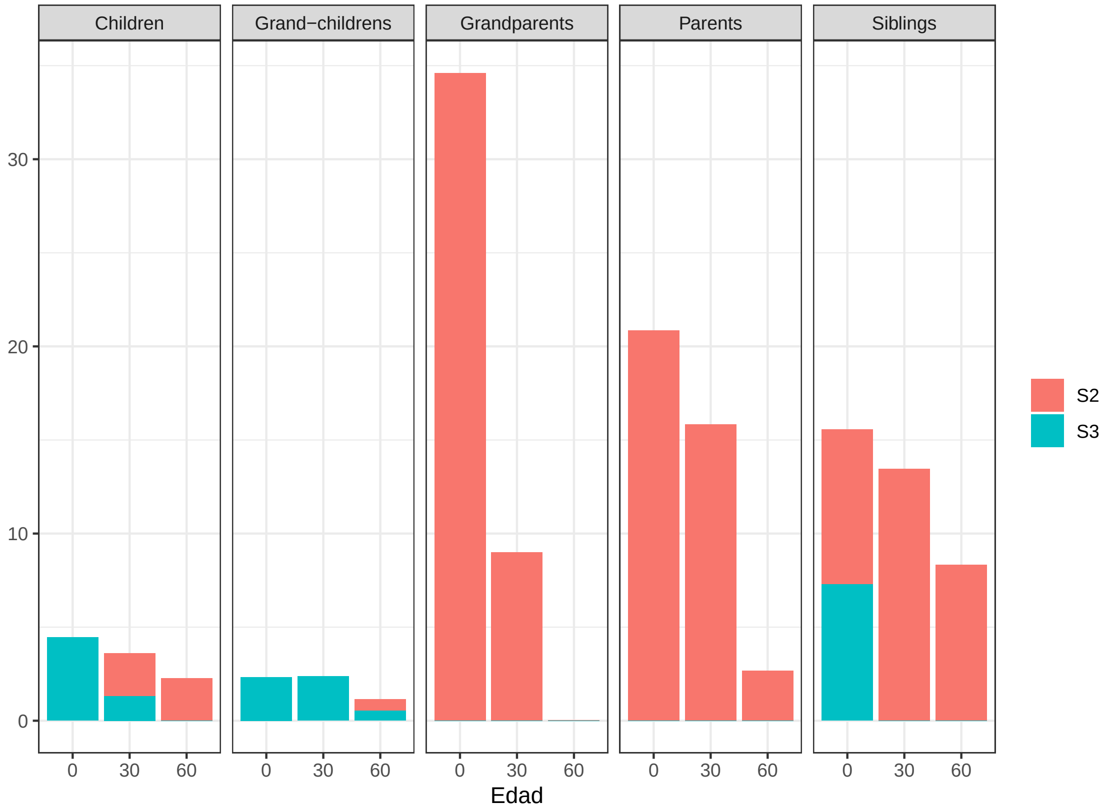

Figure 5: Shared kin-years lost for selected kin types in Argentina 2015. Decomposing $S(x)$ in analogy with $T(x)$ indicator: $S2$ represents those kin-years for population born after Ego´s age and $S3$ for those that would born after (grandparents and parents are all $S2$).

This analysis gets more interesting when comparing two populations in different stages in demographic transition, to inspect how changing mortality and fertility can change bereavement average experience. Let´s take a look to Argentina and Guatemala in 1950 and 2015.

3.2 Argentina and Guatemala: 1950 and 2015

Given a different demographic evolution during the period 1950-2015, described in section 2.1, Argentina and Guatemala also faced a different transition in kinship size and structure, and in consequence in bereavement patterns. Many conclusions can be obtained from figure 6, in terms of gaps in mean bereavement experience. Regarding children (first column of the plot), the overlapping experience (dying children before parents) has a considerable reduction for both countries at age 30, influenced by less amount of children. The amount of kin-years without children is more than double for Guatemala compared to Argentina, being 37 over 40 of $e(0)$, almost 100%. The unexpected children deaths at age 60 reduces in a similar way from 93% to 85% in Argentina, and from 95% to 85% for Guatemala.

From the grandparents side (second column of the plot), there is an increase of loss of shared time (doubling for Guatemala) until same values round 34 kin-years for both countries in 2015, as

Age	e(x)	Kin	L(x)	D(x)	P(x)	S(x)	M(x)	U(x)	O(x)
0	80.62	Children	0.00	0.00	NaN	4.45	0.00	0.00	0.00
0	80.62	Grandparents	3.65	0.03	0.00	34.62	0.00	0.00	0.00
0	80.62	Parents	2.00	0.00	0.00	20.84	0.00	0.00	0.00
0	80.62	Siblings	1.06	0.00	0.00	15.56	0.00	0.00	0.00
30	51.99	Children	1.37	0.02	0.01	3.62	0.21	1.00	0.08
30	51.99	Grandparents	1.51	2.24	0.59	9.01	0.43	0.00	0.31
30	51.99	Parents	1.81	0.21	0.10	15.84	0.33	0.03	0.08
30	51.99	Siblings	2.23	0.05	0.02	13.45	0.48	0.78	0.19
60	24.35	Children	2.23	0.06	0.03	2.26	0.34	0.85	0.04
60	24.35	Grandparents	0.01	3.64	1.00	0.03	0.56	0.00	0.26
60	24.35	Parents	0.59	1.46	0.70	2.68	0.28	0.00	0.19
60	24.35	Siblings	1.95	0.34	0.14	8.35	0.25	0.13	0.18

Table 1: Bereavement indicators by age and type of kin. Argentina 2015. L(x) living kin, D(x) already death kin, S(x) expected loss of shared years, O(x) overlapping experience, U(x) unexpected deaths assuming the age of 30 as threshold. See appendix for definition on M(x) proportion of life since lost and P(x) proportion of deaths over all met.

a product of increasing survival: Focal has more shared years to lose rather when kin dies sooner.

Probably the most remarkable result for parents is the decrease of overlapping experience of parents compared to grandparents, sharing more than a quarter of the death experience to the half (round 12%), see figure 7). In terms of mean age at death, both increased for parents and grandparents, but more for the first ones, decreasing the overlapped experience.

For siblings, there is a considerable reduction for Argentina in overlapping experienced deaths between siblings and parents, from 31% to 18%, but not for Guatemala. A Guatemalan woman expected to spend more than 40 kin-years in bereavement because of her siblings, with a life expectancy round this value too, an historical disadvantage in terms of burden of bereavement.

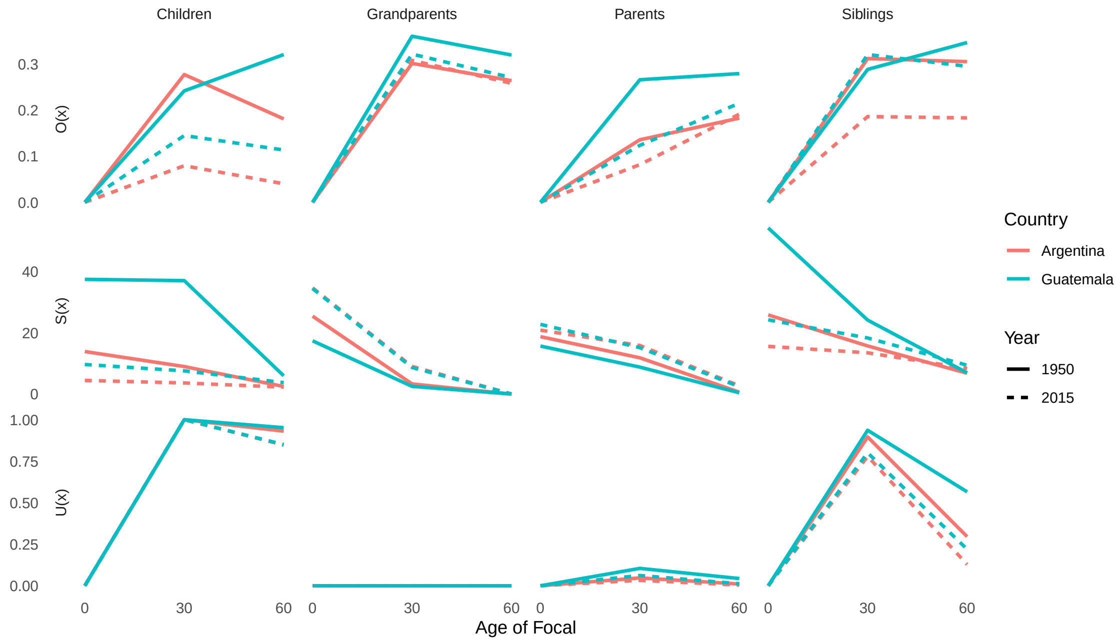

Figure 6: bereavement indicators for Guatemala and Argentina in 1950 and 2015. See 2.

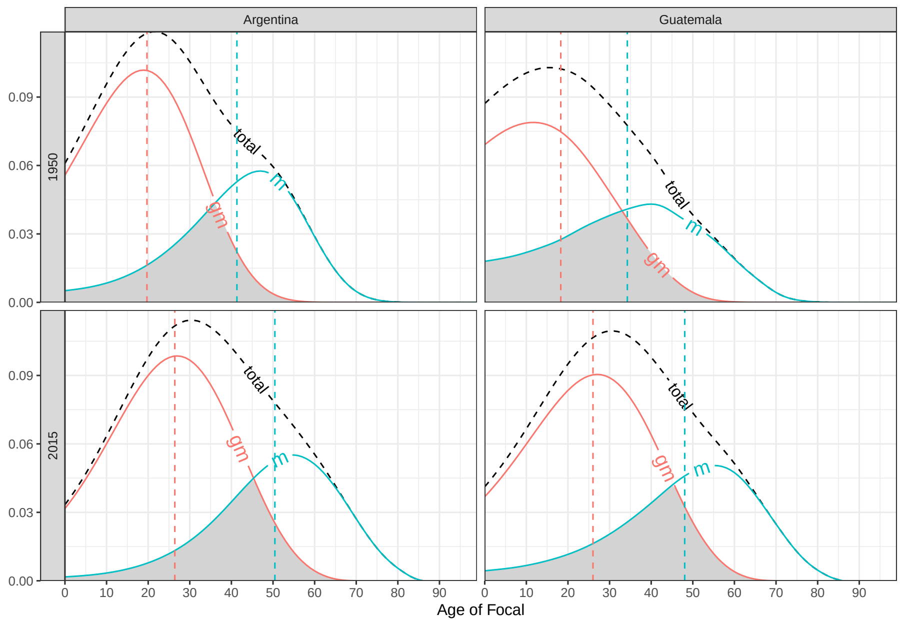

Figure 7: Count distribution of deaths of parents and grandparents, by age of Focal, and mean age of experience for each kin (dashed line). Overlapping experience at some age $O(x)$ is the grey area over total until that same age x. Argentina and Guatemala, 1950 and 2015).

4 Discussion

Grief impact was proven to be dependent on the type of kin, which suggests that a score can be used to construct a weighted value at each age for the indicators in this work (Fernández-Alcántara and Zech, 2017). In that context, a grief scale can be interpreted in a multi-state framework, with successive stages (Caswell, 2020), and this approach can be decomposed within grief steps. Here we do not include this: we avoid any comment about if losing 0.1 children carries more grief or burden than losing 1.5 aunts. In some cases, grief indicators all peak within approximately 6 months post-loss. Those who score high on these indicators beyond 6 months post-loss might benefit from further evaluation for prolonged grief disorder (Paul K. Maciejewski, 2007; Prigerson et al., 2009). Other complexities that was tried to consider was the timing of the loss in the life-course, and how was that loss in terms of potential grief (unexpected, overlapped). This work only gives some indicators to infer which characteristics, at the population-level, can be compared in time and place, to orient policy interventions.

The concept of kin-years is not so straightforward to handle: what means having 10 years of bereavement for one child and 5 for another, how those can be summed up or combined? This kind of analysis shows the need of having some multidisciplinary discussion in this topics. Also, how can be related those years with Ego´s expected years of life, even when they are the same quantity.

Many measures can be prospective or retrospective, and formulas can be changed in that sense, looking what Ego must affront or what already has passed in terms of kin loss. This would depend on research interest.

The overlapping kin loss is a rough indicator, not focused in giving a probability but in the share of the death experience from Ego´s view. Doing microsimulation turns easier to compute the percentage of people that, at certain age, had lost a grandchildren earlier than a children. An additional advantage there is the possibility of measuring uncertainty. Dispersion over the mean in all indicators is critical in mental health policies: the tails are the priority.

5 Appendix

5.1 Figures

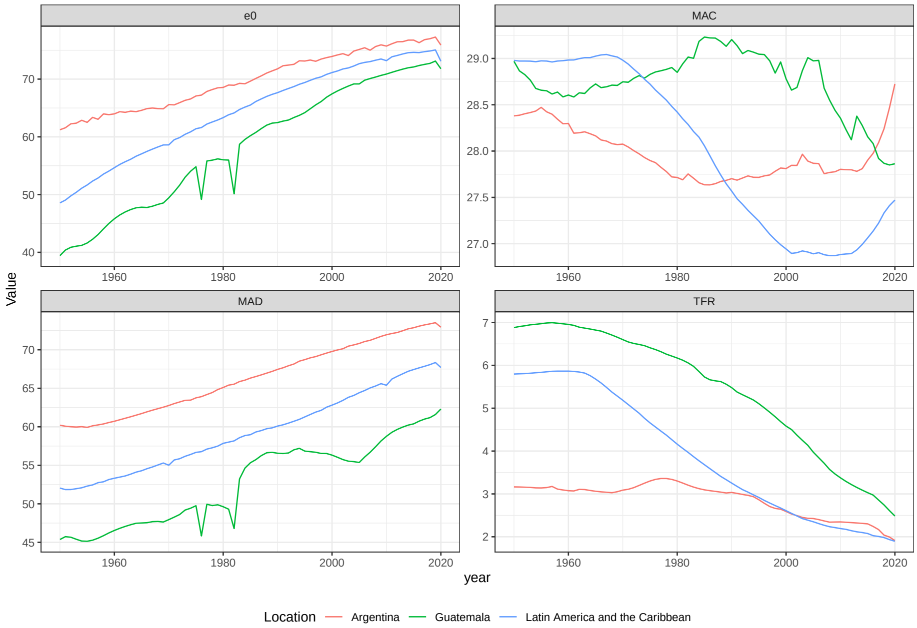

Figure 8: Level and mean age for mortality and fertility in Guatemala and Argentina. years between 1950 and 2015

5.2 Tables

5.3 Other indicators

5.3.1 Proportion of life since lost

Determine the mean portion of life where Focal had been in bereavement because of kin type k. Being $\overline{a}(k)$ the mean age at loss until age x, then:

$$\mathsf{M}(x) = \int_0^x \frac{d(y)}{D(x)} \frac{(x-y)}{x}\, dy = 1 - \frac{\overline{a}(x)}{x} \tag{8}$$

Compared to the previous, this is a retrospective indicator, measured in a portion of years lived, conditional on Focal being alive at age x. It is not a measure of intensity, given that kin types with very rare deaths until Focal is aged x (for example, great-granddaughters) will always have a mean time of occurrence, but actually with a very low count.

Death proportion over kin ever met

We know how many total relatives, on average, Focal has lost when she is x, but probably what matters for her grief experience is the relationship of that quantity with the amount of kin ever met by her Caswell et al. (2023). Being $L(x)$ the total kin ever met by Focal (since she was born), this portion is:

$$\mathsf{P}(x) = D(x)/L(x) \tag{9}$$

This is a retrospective indicator, measuring the portion died of the total ever lived since Focal was born. This last function $L(x)$ is obtained by projecting population kin but assuming no mortality risk, only for the specific kin type.

5.3.2 Age of more death experience

Given the function $d(x)$, find the age that maximizes the number of death kin over the rest of Focal's life:

$$\mathsf{H}(x) = \max_{y \geq x} d(y) \tag{10}$$

For ancestors, the problem is related to finding the modal age at death from the distribution $d(x)$ in the life table, because no new individuals arrive. This depends on the age distribution of ancestors when Focal was born: great-grandparents could have a mean age higher than the modal age, the value is always the first year of Focal's future life. In the case of kin types that increases its size with Focal´s age, it depends on fertility patterns and the time sequence of generations.

5.3.3 Mean size of potential support network

A new relative death can find Focal surrounded by a big network or alone. The abundance of living kin is important for contention and sharing loss situations (like economics). The age when Focal has fewer functional living kin is relevant. We define the support group for a loss of type k when Focal is x, $\overline{k}_A(x)$, as the group of living kin of certain types, with ages over A. In that context, the age when Focal has fewer living kin to be supported for that loss can be found in these terms:

$$\mathsf{N}(x) = \min_{y \geq x} \overline{k}_A(y) \tag{11}$$

For example, if the kin of interest is daughters, and the support net includes sisters, mothers, and aunts between 20 and 80 years old, for the case of Argentina in 1960, the ages in the sixties are those where almost no mothers and aunts are left, sisters started to decline because of mortality, and granddaughters starts to be born.

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Country	Year	Age	e(x)	Kin	L(x)	D(x)	P(x)	S(x)	M(x)	U(x)	O(x)
Argentina	1950	0	64.83	Children	0.00	0.00	NaN	13.86	0.00	0.00	0.00
Argentina	2015	0	80.62	Children	0.00	0.00	NaN	4.45	0.00	0.00	0.00
Argentina	1950	0	64.83	Grandparents	3.27	0.06	0.00	25.40	0.00	0.00	0.00
Argentina	2015	0	80.62	Grandparents	3.65	0.03	0.00	34.62	0.00	0.00	0.00
Argentina	1950	0	64.83	Parents	2.00	0.01	0.00	18.72	0.00	0.00	0.00
Argentina	2015	0	80.62	Parents	2.00	0.00	0.00	20.84	0.00	0.00	0.00
Argentina	1950	0	64.83	Siblings	1.31	0.01	0.00	25.80	0.00	0.00	0.00
Argentina	2015	0	80.62	Siblings	1.06	0.00	0.00	15.56	0.00	0.00	0.00
Argentina	1950	30	42.29	Children	1.82	0.17	0.08	8.97	0.19	1.00	0.28
Argentina	2015	30	51.99	Children	1.37	0.02	0.01	3.62	0.21	1.00	0.08
Argentina	1950	30	42.29	Grandparents	0.68	2.67	0.79	3.31	0.49	0.00	0.30
Argentina	2015	30	51.99	Grandparents	1.51	2.24	0.59	9.01	0.43	0.00	0.31
Argentina	1950	30	42.29	Parents	1.59	0.44	0.20	11.82	0.37	0.05	0.14
Argentina	2015	30	51.99	Parents	1.81	0.21	0.10	15.84	0.33	0.03	0.08
Argentina	1950	30	42.29	Siblings	2.73	0.28	0.09	15.69	0.59	0.90	0.31
Argentina	2015	30	51.99	Siblings	2.23	0.05	0.02	13.45	0.48	0.78	0.19
Argentina	1950	60	17.10	Children	2.75	0.41	0.13	2.47	0.41	0.93	0.18
Argentina	2015	60	24.35	Children	2.23	0.06	0.03	2.26	0.34	0.85	0.04
Argentina	1950	60	17.10	Grandparents	0.00	3.27	1.00	0.00	0.67	0.00	0.26
Argentina	2015	60	24.35	Grandparents	0.01	3.64	1.00	0.03	0.56	0.00	0.26
Argentina	1950	60	17.10	Parents	0.18	1.85	0.91	0.63	0.35	0.01	0.18
Argentina	2015	60	24.35	Parents	0.59	1.46	0.70	2.68	0.28	0.00	0.19
Argentina	1950	60	17.10	Siblings	2.09	0.97	0.31	6.80	0.37	0.30	0.31
Argentina	2015	60	24.35	Siblings	1.95	0.34	0.14	8.35	0.25	0.13	0.18
Guatemala	1950	0	40.32	Children	0.00	0.00	NaN	37.41	0.00	0.00	0.00
Guatemala	2015	0	75.62	Children	0.00	0.00	NaN	9.64	0.00	0.00	0.00
Guatemala	1950	0	40.32	Grandparents	2.65	0.07	0.00	17.36	0.00	0.00	0.00
Guatemala	2015	0	75.62	Grandparents	3.48	0.04	0.00	34.39	0.00	0.00	0.00
Guatemala	1950	0	40.32	Parents	2.00	0.02	0.00	15.65	0.00	0.00	0.00
Guatemala	2015	0	75.62	Parents	2.00	0.00	0.00	22.72	0.00	0.00	0.00
Guatemala	1950	0	40.32	Siblings	2.09	0.08	0.00	54.16	0.00	0.00	0.00
Guatemala	2015	0	75.62	Siblings	1.32	0.00	0.00	24.20	0.00	0.00	0.00
Guatemala	1950	30	33.18	Children	2.93	1.07	0.25	36.99	0.20	1.00	0.24
Guatemala	2015	30	48.62	Children	1.80	0.05	0.03	7.57	0.21	1.00	0.15
Guatemala	1950	30	33.18	Grandparents	0.51	2.19	0.81	2.52	0.54	0.00	0.36
Guatemala	2015	30	48.62	Grandparents	1.42	2.15	0.59	8.65	0.44	0.00	0.32
Guatemala	1950	30	33.18	Parents	1.22	0.82	0.39	8.79	0.45	0.10	0.27
Guatemala	2015	30	48.62	Parents	1.71	0.31	0.15	15.13	0.39	0.06	0.12
Guatemala	1950	30	33.18	Siblings	3.89	1.95	0.33	24.12	0.59	0.94	0.29
Guatemala	2015	30	48.62	Siblings	2.81	0.14	0.05	18.29	0.48	0.80	0.32
Guatemala	1950	60	13.88	Children	4.00	2.88	0.42	5.96	0.41	0.95	0.32
Guatemala	2015	60	22.32	Children	2.82	0.19	0.06	3.74	0.33	0.85	0.11
Guatemala	1950	60	13.88	Grandparents	0.00	2.65	1.00	0.00	0.70	0.00	0.32
Guatemala	2015	60	22.32	Grandparents	0.01	3.47	1.00	0.03	0.57	0.00	0.27
Guatemala	1950	60	13.88	Parents	0.14	1.88	0.93	0.42	0.47	0.04	0.28
Guatemala	2015	60	22.32	Parents	0.54	1.51	0.73	2.41	0.32	0.01	0.22
Guatemala	1950	60	13.88	Siblings	2.32	3.57	0.60	6.92	0.54	0.57	0.35
Guatemala	2015	60	22.32	Siblings	2.34	0.64	0.21	9.40	0.32	0.22	0.30

Table 2: Bereavement indicators by age and type of kin. Argentina and Guatemala 1950 and 2015